

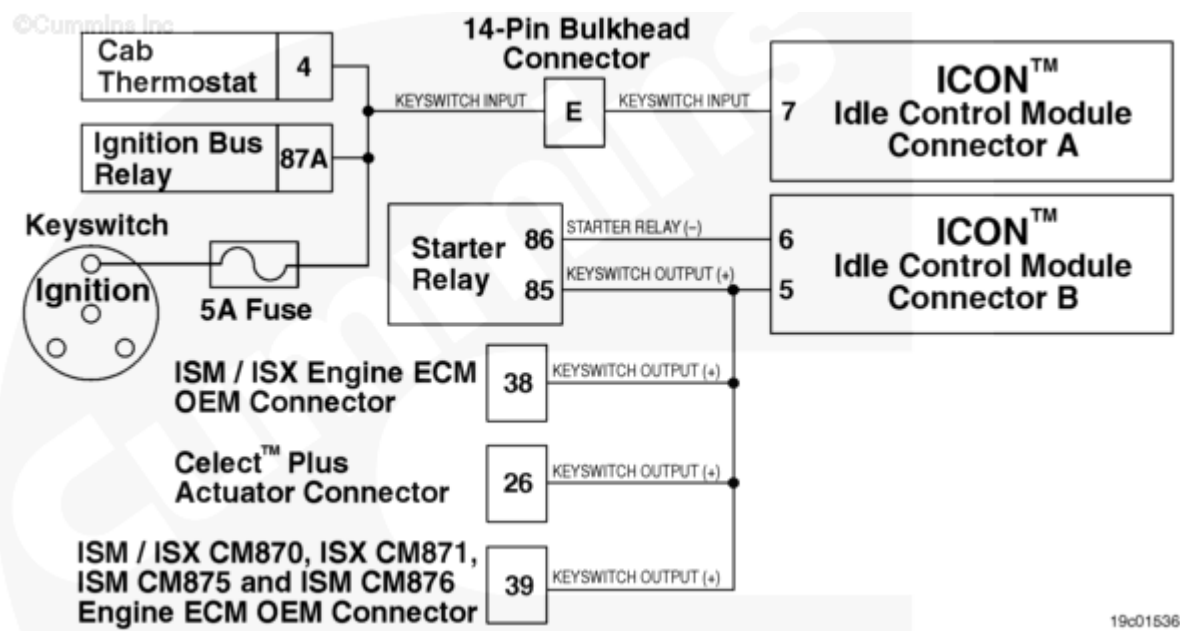
## Fault Code: 587 (Aftermarket and OEM)

### Keyswitch Circuit - Voltage Below Normal or Shorted to Low Source

 Printable Version

#### Overview

| Codes                                                                                  | Reason                                                                                                                                                                                                                    | Effect                                                                                                                                   |
|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Fault Code: 587</p> <p>PID(P):</p> <p>SPN:</p> <p>FMI:</p> <p>Lamp:</p> <p>SRT:</p> | Keyswitch Circuit - Voltage Below Normal or Shorted to Low Source. Low voltage detected at the keyswitch output circuit of the ICON™ idle control module when high voltage was expected by the ICON™ idle control module. | The ICON™ system will be disabled. Mandatory shutdown can also be disabled. Engine will <b>not</b> start normally. Engine will shutdown. |



#### Circuit Description

The keyswitch output circuit provides the keyswitch input signal for the engine control module and starter relay. The above circuit diagram can vary, such as connector or pins, depending on the vehicle make or model. OEM installations can possibly provide the harnessing between the idle control module and other ICON™ devices.

## Component Location

The keyswitch signal connects to pin 26 of the actuator connector on the engine harness for SELECT™ Plus engines, and pin 39 for the CM870 and CM875 electronic control module (ECMs), and pin 38 of the OEM connector for ISM, ISX, and Signature engines. It also connects to pin 85 of the starter relay. This assumes the keyswitch input voltage from the keyswitch to pin 7 of the ICON™ idle control module A connector is present.

The ICON™ module can be located in a different location depending on the vehicle application.

## Shoptalk

This fault typically indicates a short circuit to ground. The engine will possibly **not** be able to start.

The ICON™ system can display **only** the present active fault code. If more than one fault code is active at the same time, the ICON system flashes out the highest priority fault. After the fault has been corrected then the next active fault will be flashed out.

**Note:** The ICON™ electronic service tool can display more than one active and or inactive fault codes at the same time.

## Warnings and Cautions

### CAUTION

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement:

Part Number 3822758 - male Deutsch/AMP/Metri-Pack test lead

Part Number 3822917 - female Deutsch/AMP/Metri-Pack test lead.

## Troubleshooting Summary

| STEPS   |                                                                                                              | SPECIFICATIONS        |
|---------|--------------------------------------------------------------------------------------------------------------|-----------------------|
| STEP 1. | Read the fault codes.                                                                                        |                       |
|         | <b>STEP 1A.</b> Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes. | Fault Code 587 active |
| STEP 2. | Check the ICON™ idle control module.                                                                         |                       |

| STEPS          |                                                                                                | SPECIFICATIONS         |
|----------------|------------------------------------------------------------------------------------------------|------------------------|
|                | <b>STEP 2A.</b> Inspect the ICON™ engine harness and ICON™ idle control module connector pins. | No damaged pins        |
|                | <b>STEP 2B.</b> Check the output of the ICON™ idle control module.                             | Greater than 10 VDC    |
| <b>STEP 3.</b> | Check the ICON™ harness.                                                                       |                        |
|                | <b>STEP 3A.</b> Check for a short circuit to ground.                                           | More than 100k ohms    |
|                | <b>STEP 3A-1.</b> Check the engine ECM.                                                        | Fault becomes inactive |
| <b>STEP 4.</b> | Clear the fault code.                                                                          |                        |
|                | <b>STEP 4A.</b> Disable the fault code.                                                        | Fault Code 587 cleared |

## STEP 1. Read the fault codes.

**STEP 1A. Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes.**

### Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect the ICON™ electronic service tool.

| Action                                                                                       | Specification/Repair                                                                                                  | Next Step |
|----------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------|
| Use the fault flashout feature or the ICON™ electronic service tool to read the fault codes. | Fault Code 587 active.                                                                                                | <b>2A</b> |
|                                                                                              | Refer to Inactive or Intermittent Fault Codes, Procedure 019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html). | <b>4A</b> |

## STEP 2. Check the ICON™ idle control module.

### STEP 2A. Check the ICON™ idle control module connector pins.

#### Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.

| Action                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Specification/Repair                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Next Step |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Inspect the ICON™ idle control module connector pins for the following: <ul style="list-style-type: none"><li>• Loose connector</li><li>• Corroded pins</li><li>• Bent or broken pins</li><li>• Pushed back or expanded pins</li><li>• Moisture in or on the connector</li><li>• Missing or damaged connector seals</li><li>• Dirt or debris on or in the connector pins</li><li>• Connector shell broken</li><li>• Wire insulation damage</li><li>• Damaged connector locking tab.</li></ul> For general inspection techniques, refer to Component Connector and Pin Inspection, Procedure 019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html). | No damaged pins                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>2B</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Repair the damaged pins. <ul style="list-style-type: none"><li>• Flush the dirt, debris, or moisture from the connector pins using electrical contact cleaner, Part Number 3824510.</li><li>• Install the appropriate connector seal if it is damaged or missing.</li><li>• Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html).</li><li>• Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).</li><li>• Repair or replace the OEM wiring harness as necessary.</li></ul> | <b>4A</b> |

## STEP 2B. Check the output of the ICON™ idle control module.

### Conditions :

- Turn keyswitch ON.
- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.

| Action                                                                                                                                                                                                                         | Specification/Repair                                                                                                          | Next Step              |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Measure the voltage at pin 5 of the ICON™ idle control module B connector to engine block ground.                                                                                                                              | Greater than 10 VDC                                                                                                           | <b>3A</b>              |
| Refer to the wiring diagram or the circuit diagram for connector pin identification.<br><br>For multimeter usage techniques, refer to Multimeter Usage, Procedure 019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html). | Replace the ICON™ idle control module.<br><br>Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html). | <b>Repair complete</b> |

## STEP 3. Check the ICON™ harness.

### STEP 3A. Check for a short to ground in the ICON™ harness.

### Conditions :

- Turn keyswitch OFF.
- Disconnect the ICON™ idle control module B connector from the ICON™ idle control module.
- Disconnect the ECM connectors.

| Action | Specification/Repair | Next Step |
|--------|----------------------|-----------|
|--------|----------------------|-----------|

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| <p>Measure the resistance from pin 5 of the ICON™ Aftermarket OEM harness connector to engine block ground.</p> <p>Refer to the wiring diagram or the circuit diagram for connector pin identification.</p> <p>For multimeter usage techniques, refer to Multimeter Usage, Procedure 019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html).</p> | Greater than 100k ohms                                                                                                                                                                                                                                                                                                                                                                                                       | <b>3A-1</b> |
|                                                                                                                                                                                                                                                                                                                                                       | <ul style="list-style-type: none"> <li>Repair the ICON™ engine harness. Refer to Procedure 019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html) or 019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html).</li> <li>Replace the ICON™ engine harness. Refer to Procedure 019-043 (/qs3/pubsys2/xml/en/procedures/97/97-019-043.html).</li> <li>Repair or replace the OEM wiring harness as necessary.</li> </ul> | <b>4A</b>   |

### STEP 3A-1. Check the engine fault code status.

#### Conditions :

- Turn keyswitch ON.
- Connect the ICON™ electronic service tool.
- Disconnect the engine ECM connectors.

| Action | Specification/Repair | Next Step |
|--------|----------------------|-----------|
|--------|----------------------|-----------|

|                                                                                                                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <p>Check the fault status.</p> <ul style="list-style-type: none"> <li>• Turn keyswitch OFF for 30 seconds.</li> <li>• Turn keyswitch ON.</li> <li>• Check the fault code status.</li> </ul> | <p>Fault becomes inactive</p> <p>Troubleshoot the engine ECM fault codes.</p> <ul style="list-style-type: none"> <li>• Refer to Troubleshooting and Repair Manual, CELECT™ Plus, Bulletin 3666130 (<a href="/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666130.html">/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666130.html</a>), or</li> <li>• Troubleshooting and Repair Manual, Electronic Control System, ISM, Bulletin 3666266 (<a href="/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666266.html">/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666266.html</a>), or</li> <li>• Troubleshooting and Repair Manual, Electronic Control System, Signature and ISX, Bulletin 3666259 (<a href="/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666259.html">/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/3666259.html</a>), or</li> <li>• Troubleshooting and Repair Manual, Electronic Control System, CM870 ISM, Bulletin 4021381 (<a href="/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021381.html">/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021381.html</a>), or</li> </ul> | <p><b>4A</b></p> |
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|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
|  | <ul style="list-style-type: none"> <li>• Troubleshooting and Repair Manual, Electronic Control System, CM870 Signature and ISX, Bulletin 4021334 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021334.html), or</li> <li>• Troubleshooting and Repair Manual, Electronic Control System, CM875 ISM, Bulletin 4021477 (/qs3/portal/manualviewer.html?path=/qs3/pubsys2/xml/en/outlines/4021477.html).</li> </ul> |                        |
|  | <p>Replace the ICON™ idle control module.</p> <p>Refer to Procedure 019-358 (/qs3/pubsys2/xml/en/procedures/97/97-019-358.html).</p>                                                                                                                                                                                                                                                                                                | <b>Repair complete</b> |

## STEP 4. Clear the fault code.

### STEP 4A. Disable the fault code.

#### Conditions :

- Connect all components.
- Turn keyswitch ON.

| Action | Specification/Repair | Next Step |
|--------|----------------------|-----------|
|--------|----------------------|-----------|



|                                                                                                                                            |                                                |                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|-------------------------------------------|
| Clear the inactive fault code. <ul style="list-style-type: none"><li>• Cycle the keyswitch to verify the fault code is inactive.</li></ul> | Fault Code 587 cleared                         | <b>Repair complete</b>                    |
|                                                                                                                                            | Troubleshoot any remaining active fault codes. | <b>Appropriate troubleshooting charts</b> |